HAT-P-20b

R-1389

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-20_170107` · created 2026-08-01T02:45:15+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457757.84713 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.054
Transit depth (Rp/Rs)^2	3.32 +/- 1.98 [%]
Orbital Inclination (inc)	86.79 +/- 2.42
Airmass coefficient 1 (a1)	158.64 +/- 29.78
Airmass coefficient 2 (a2)	-2.85 +/- 1.07
Scatter in the residuals of the lightcurve fit is	16.58 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	6.5
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.054 vs 0.1284 (deviation 0.99σ)
Duration fit vs published	0.0 d vs 0.0775 d (deviation infσ)
Mid-time O–C	-4142.0 min
Depth SNR	3.4
Residual scatter	16.58 %

Light curve

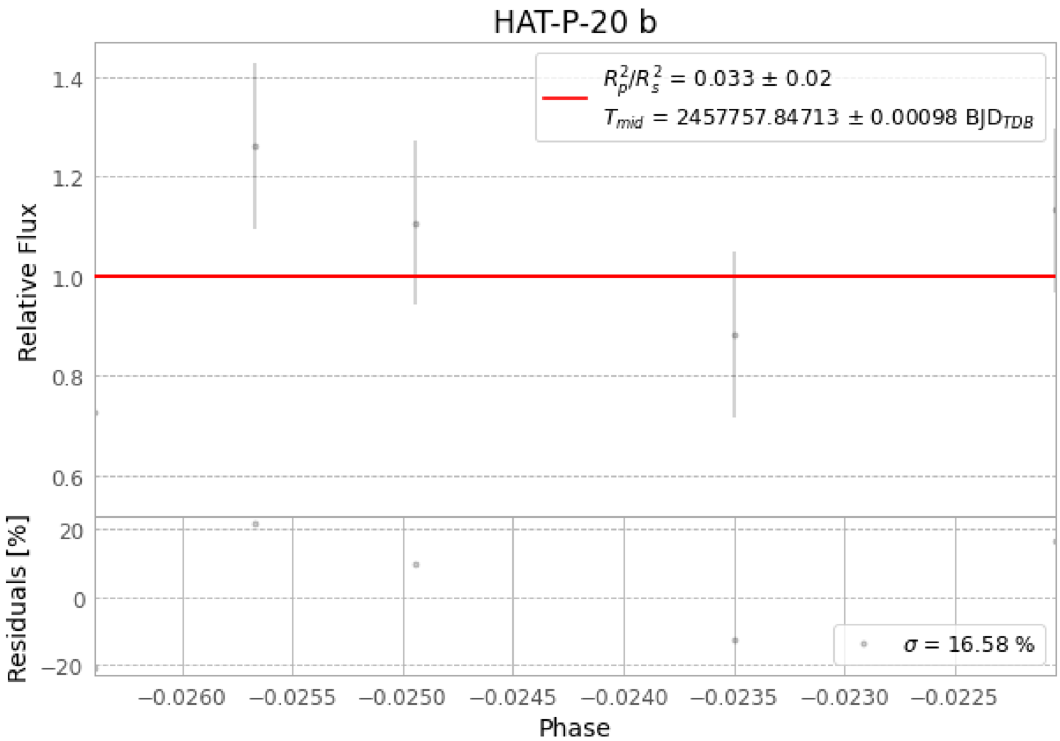

Figure 1 — FinalLightCurve_HAT-P-20 b_2017-01-06.png (SHA-256 a42ffe0013e04307…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [633, 126], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 874a62b1c993989a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

104 FITS frames (68 MB), `HATP-20170107030612.FITS` → `HATP-20170107081512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-170107.log (ddb29f1438a0…), FinalLightCurve_HAT-P-20 b_2017-01-06.png (a42ffe0013e0…), FinalLightCurve_HAT-P-20 b_2017-01-06.csv (9a082267cf33…)

Compute resources

Wall time 85.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

